

Happiness and farm productivity: insights from maize farmers in China

Happiness and
farm
productivity

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Abstract

Purpose – The purpose of this study is to explore the relationship between farmers' happiness and farm productivity, taking maize production in China as an example.

Design/methodology/approach – The conditional mixed process model is employed to account for the endogeneity that inevitably arises in regression models studying happiness and estimate the 2015 China Household Finance Survey data.

Findings – The empirical results show that a higher level of farmers' happiness is associated with higher maize productivity. The marginal return in maize productivity changes unevenly with increments in farmers' self-reported happiness on a five-point Likert scale. Farmers' happiness is positively determined by their age, educational level, farm size, machinery ownership, access to agricultural subsidy and car ownership. Machinery ownership and access to credit are two important factors that improve maize productivity.

Research limitations/implications – The finding suggests that promoting the subjective well-being of farmers is conducive to higher productivity and improved national food security. The results have implications for China and other developing countries aiming for sustainable agricultural development.

Originality/value – Firm-level data show that workers' happiness improves productivity. However, it is still unclear whether farmworkers' happiness affects farm productivity. Thus, this study provides the first empirical analysis of the impact of farmworkers' happiness on farm productivity.

Keywords Happiness, Productivity, Maize farmers, CMP model

Paper type Research Paper

1. Introduction

Does “happiness” make workers more productive? This is the principal question that we address in this paper. Investigating the linkage between workers' well-being and productivity (e.g. labor productivity or total factor productivity) has become increasingly popular as improving productivity is the key to economic growth (DiMaria *et al.*, 2020; Isham *et al.*, 2021). Subjective well-being has attracted attention because it is a valuable alternative to monitoring people's quality of life beyond traditional economic measures such as income and expenditure (Zhu *et al.*, 2021). It reflects peoples' satisfaction with their overall lives and specific aspects of their lives, such as occupations and social connections.

The happy productive worker theory suggests that improvements in workers' well-being generate productivity gains. Previous studies have reported empirical evidence corroborating this theory by analyzing data collected from firm workers (Bryson *et al.*, 2017; DiMaria *et al.*, 2020) or designing laboratory experiments (Oswald *et al.*, 2015). More specifically, Oswald *et al.* (2015) showed that happier individuals report around 12%



JEL Classification — D24, Q12, I31

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greater productivity. DiMaria *et al.* (2020) analyzed a sample of 20 European countries and found that life satisfaction significantly increases productivity at the individual and firm levels. In their review on worker well-being and productivity in advanced economies, Isham *et al.* (2021) emphasized that high job satisfaction is linked with higher individual and organizational productivity, but sometimes, the positive relationship is affected by resource-intensive mediators.

The link between happiness and productivity is affected by the context in which people operate. Thus, researchers have studied these links for various populations in different settings. For example, Proto *et al.* (2012) analyzed university students' subjective well-being and productivity using laboratory tasks. DiMaria *et al.* (2020) examined macroeconomic data on 20 European countries to examine productivity gains from improved subjective well-being, and Bellet *et al.* (2019) studied how happiness influences the productivity of British Telecom employees. Notwithstanding the broadly consistent results across these studies, researchers recognize and acknowledge that their findings are relevant to specific contexts – generalizations may be misleading. With this in mind, the present study focuses on the happiness productivity nexus for farmers in rural China.

Working on farms is notably different from working in corporate workplaces. Farmers work in rural areas, whereas corporate jobs are located in cities. Living in remote areas lacking basic amenities, farmers can feel socially and geographically isolated. Farmers produce tangible goods and are intimately involved with the overall production process – this is gratifying and leads to greater happiness (Treleaven, 2012). In contrast, corporate workers tend to have a narrow focus on their specialty and may be disconnected from and oblivious to the end-to-end production process. Although corporate workers may enjoy steadier incomes, farmers benefit from agricultural subsidies. Visual exposure to gloomy weather, which is known to affect happiness and this productivity, may affect farmers more than corporate employees (Bellet *et al.*, 2019). Farmers are self-employed and produce subsistence and commercial crops, whereas corporate employees supply labor in exchange for wages. Thus, farmers may be more vested in the outcome of their efforts; they may work productively despite experiencing unhappiness. Farmers are influenced by factors largely out of their control: commodity prices, weather and interest rates. It bears emphasis that farmers are more vulnerable than the general population to suicide. Thus, results derived from the analyses of corporate workers may not be generalized to farmers.

This paper extends the literature on subjective well-being and productivity by providing the first empirical analysis of the impact of farmers' happiness on farm productivity. This issue has been overlooked in the literature; however, it is worthy of attention for at least two reasons. First, the United Nations' Sustainable Development Goals emphasize the importance of ensuring healthy lives and promoting well-being for all ages (goal 4), and farmers are no exception. Nevertheless, there is a paucity of research on rural farmers' well-being (Nie *et al.*, 2020; Zhu *et al.*, 2021). Second, farmers are an important social group, and their farm productivity performance plays a pivotal role in ensuring national and global food security and boosting economic growth. Furthermore, constituting significant proportions of the developing countries' populations, farmers play a pivotal role in their nations' economic growth. For example, 39.4% of the population lives in rural areas of China in 2019, contributing to around 7% of the country's total GDP (NBS, 2020). Considering the scale and scope of farming in developing countries, it is surprising that the nexus between farmers' happiness and farm productivity has not received due attention. This paper fills this critical void in the literature.

We analyze open-access 2015 China Household Finance Survey data, using maize production as an example. In 2019, harvesting 41.3 million ha of land, China produced 261 million tons of maize – globally, it ranked first and second in terms of area harvested and production, respectively (FAOSTAT). However, maize productivity per hectare of land was ranked only 50th globally, undermining China's food security and the United Nations' sustainable development goals of “zero hunger (Goal 2)” and “responsible consumption and

production (Goal 12)". We employ a conditional mixed process model to conduct the empirical analysis. The model allows us to estimate the happiness and production equations jointly while accounting for the potential endogeneity of the happiness variable.

The remainder of the paper is organized as follows. [Section 2](#) discusses the methods and data used in this paper. The empirical results are presented in [Section 3](#). The final section concludes.

2. Methodology

2.1 Conditional mixed process model

We investigate whether farmers' happiness increases farm productivity. Happiness influences and is influenced by many factors. Thus, endogeneity is likely to arise in regression models of happiness ([Katsaiti, 2012; Zhu et al., 2021](#)). Also, happiness is measured as an ordinal variable, and maize productivity is a continuous variable. In light of these two reasons, we employ a conditional mixed process (CMP) model to study the link between farmers' happiness and productivity ([David, 2011](#)). The CMP model jointly estimates one ordered probit model for the happiness equation and one ordinary least square (OLS) regression model for the maize productivity equation as follows:

$$H_i^* = X_i\alpha_i + IV_i\beta_i + \varepsilon_i, \text{ with } H_i = \begin{cases} 1 & \text{if } H_i^* \leq O_1 \\ 2 & \text{if } O_1 < H_i^* \leq O_2 \\ \dots & \dots \\ J & \text{if } O_{J-1} \leq H_i^* \end{cases} \quad (1)$$

$$Y_i = H_i\delta_i + X_i\eta_i + \vartheta_i \quad (2)$$

where H_i^* is a latent variable, which reflects farmer i 's happiness level and is represented by an observed category variable H_i . Particularly, H_i is determined by the unknown cutoffs $O_1, O_2, \dots, O_{J-1}$: these satisfy the condition $O_1 < O_2 < \dots < O_{J-1}$. X_i is a vector of control variables (e.g. farmers' age and education, household size and farm size). IV_i refers to the instrumental variable (IV) employed to identify the CMP model; Y_i refers to maize productivity, measured at kg/mu; $\alpha_i, \beta_i, \delta_i$ and η_i are parameters to be estimated and ε_i and ϑ_i are two error terms, capturing the unobserved heterogeneities.

The CMP model estimates [Equations \(1\) and \(2\)](#) jointly using the maximum likelihood estimator and generates a correlation term $\rho_{\varepsilon\vartheta} = \text{corr}(\varepsilon_i, \vartheta_i)$. A statistically significant $\rho_{\varepsilon\vartheta}$ suggests that happiness is an endogenous variable and confirms the effectiveness of the CMP model ([David, 2011](#)). This paper employs farmers' donation status as the IV in [Equation \(1\)](#). The donation variable is given a value of one if a farmer reports that he or she has donated to others and zero otherwise. As donation is a way to voluntarily contribute to society, it may positively affect farmers' happiness. The existing studies have shown that engaging in prosocial activities such as charitable giving and voluntary donation promotes people's positive emotions, including happiness and life satisfaction ([Aknin and Whillans, 2021; Thoits and Hewitt, 2001](#)). However, donation behavior does not directly impact maize productivity. Thus, donation logically meets the requirement of a valid IV. To confirm these results, we use Pearson correlation analysis and a falsification test to check the validity and effectiveness of the selected IV. The Pearson correlation analysis (see the upper part of [Table A1](#) in the [Appendix](#)) shows that donation is significantly correlated with farmers' happiness but not with maize productivity. The results of the falsification test (see the lower part of [Table A1](#)) suggest that donation has a statistically significant impact on farmers' happiness, but it does not have a significant impact on maize productivity, even at the

10% significance level. The findings unanimously confirm the appropriateness of the employed IV.

2.2 Data

We use open-access data from the 2015 China Household Finance Survey (CHFS). The survey is conducted by the Survey and Research Center for China Household Finance at the Southwestern University of Finance and Economics in Chengdu, China. The CHFS is a nationally representative survey, collecting detailed information on household and individual-level characteristics (e.g. income, expenditure, psychology such as happiness level) and farm information such as land size and crop yields [1]. The 2015 CHFS reached 37,341 households (including 11,635 rural household observations) from 353 counties and 29 provinces (excluding Xinjiang and Tibet) in Mainland China, capturing household information for 2014. Given that we are only interested in the relationship between farmers' happiness and maize productivity, we follow a two-stage process to clean our data. First, we exclude urban observations and rural observations without maize production. Then, we remove observations with missing information, resulting in 4,212 valid observations for empirical analysis.

The dependent variable in this paper is maize productivity, expressed as kg/mu (1 mu = 1/15 hectare). Happiness is the key explanatory variable. In the CHFS questionnaire, the respondents were asked the following question: "In general, to what extent do you feel happy?" Respondents are asked to answer the question based on a five-point Likert scale, ranging from 1 = very unhappy, 2 = unhappy, 3 = fair, 4 = happy and 5 = very happy. We focus on the self-reported happiness level of household heads.

We draw upon the existing literature on subjective well-being and productivity to select the control variables (e.g. [Devoto et al., 2012](#); [Frijters and Beaton, 2012](#); [Gul and Pesendorfer, 2007](#); [Isham et al., 2021](#); [Kollamparambil, 2020](#); [Liu et al., 2020](#); [Narayanan, 2016](#); [Zhou et al., 2020](#)). Specially, we include age, gender, education, off-farm work, cadre, household size, farm size, land certification, machinery, subsidy, access to credit, car ownership and location dummies as the control variables. Among them, machinery is defined as a dummy variable, which equals 1 if a farm household owns machinery, such as tractor, seeder, rice transplanter, harvester, thresher, rotary cultivator, pumper or pesticide sprayers, for farm production and 0 otherwise. The subsidy variable is given a value of 1 if a household receives agricultural subsidies (money, production materials or both) in 2014 and 0 otherwise. Access to credit, too, is defined as a dummy variable, which equals 1 if a farm household obtains credit from informal sources (e.g. friends, relatives or neighbors), formal financial institutions (e.g. banks and credit cooperatives) or both for purchasing production inputs (e.g. fertilizers, pesticides and seeds) in 2014, and 0 otherwise.

[Table 1](#) presents the definitions and descriptive statistics of the selected variables. It shows that average maize productivity is 247 kg/mu (1mu = 1/15 hectare), and the mean happiness level is 3.6 on a five-point Likert scale. On average, household heads are 55 years old and have primary-to-middle school education; households have just under three members and own around 11 mu of land. Around 29% of households obtained credit to support agricultural production. Households owning cars account for only 15% of our samples.

[Figure 1](#) shows a positive association between maize productivity and household heads' happiness levels. In fact, productivity increases monotonically with greater levels of happiness. Although the findings are suggestive, they are only descriptive. Thus, to gain deeper insights, we employ the CMP model to examine the relationship between farmers' happiness and productivity.

Variables	Definition	Mean (S.D.)	Happiness and farm productivity
<i>Dependent variables</i>			
Maize productivity	Maize productivity (1,000 kg/mu) ^a	0.247 (0.150)	
Happiness	Self-reported happiness level of household head: from 1 = very unhappy to 5 = very happy	3.639 (0.860)	
<i>Independent variables</i>			
Age	Age of household head (years)	54.694 (11.062)	
Gender	1 if household head is male, 0 otherwise	0.893 (0.309)	
Education	Educational level of household head ^b	2.620 (0.960)	
Off-farm work	1 if household head worked off-farm, 0 otherwise	0.355 (0.479)	
Cadre	1 if household head is a village cadre, 0 otherwise	0.047 (0.211)	
Household size	Total number of members in household (persons)	2.763 (1.706)	
Farm size	Total farm size (mu)	10.870 (59.053)	
Land certification	1 if household has a land certificate, 0 otherwise	0.452 (0.498)	
Machinery	1 if household owns machinery, 0 otherwise	0.391 (0.488)	
Subsidy	1 if household received agricultural subsidy, 0 otherwise	0.789 (0.408)	
Access to credit	1 if household has access to credit, 0 otherwise	0.293 (0.455)	
Car	1 if household possess a car, 0 otherwise	0.147 (0.354)	
East	1 if household is located in eastern region, 0 otherwise	0.295 (0.456)	
Central	1 if household is located in central region, 0 otherwise	0.362 (0.481)	
West	1 if household is located in western region, 0 otherwise	0.344 (0.475)	
Donation (IV)	1 if household donates to others, 0 otherwise	0.798 (0.402)	
Note(s): Standard deviations are presented in parentheses			
^a 1 mu = 1/15 hectare; ^b 1 = illiteracy, 2 = primary school, 3 = middle school, 4 = high school, 5 = secondary specialized school, 6 = junior specialized school, 7 = undergraduate, 8 = master, 9 = PhD			

Table 1.

Variables definition and descriptive statistics

Table 1.
Variables definition and descriptive statistics

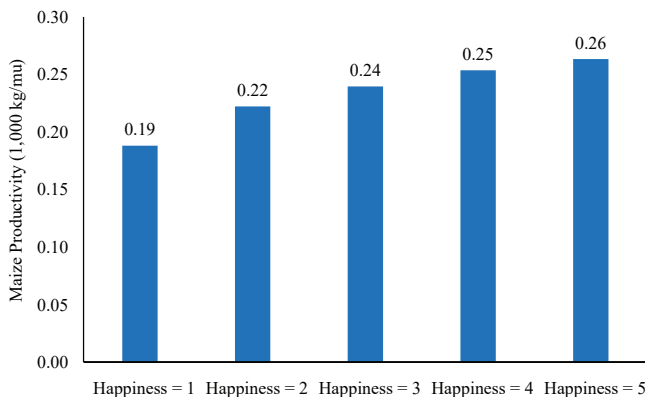


Figure 1.
Distributions of maize productivity by happiness levels

3. Results

Table 2 reports the results obtained from the CMP model. It bears emphasis that the coefficient of $\rho_{\varepsilon\theta}$ is statistically significant, suggesting that farmers' happiness is an endogenous variable – this reaffirms the value of employing the CMP model as it duly accounts for the said endogeneity. In the following two sections, we discuss the determinants of farmers' happiness and maize productivity.

Variables	Happiness	Maize productivity
Happiness		0.082 (0.023)***
Age	0.006 (0.002)***	−0.001 (0.000)***
Gender	−0.040 (0.054)	0.010 (0.008)
Education	0.038 (0.019)**	0.003 (0.003)
Off-farm work	−0.056 (0.036)	−0.006 (0.005)
Cadres	0.130 (0.079)	−0.016 (0.012)**
Household size	−0.003 (0.010)	−0.003 (0.001)**
Farm size	0.001 (0.000)**	−0.000 (0.000)
Land certification	−0.032 (0.034)	0.001 (0.005)
Machinery	0.153 (0.035)***	0.019 (0.006)***
Subsidy	0.095 (0.041)**	0.003 (0.006)
Access to credit	−0.316 (0.037)***	0.024 (0.008)***
Car	0.232 (0.049)**	0.006 (0.008)
Central	−0.102 (0.042)**	0.041 (0.006)***
West	−0.174 (0.043)***	−0.001 (0.007)
Donation (IV)	0.154 (0.039)***	
Constant		−0.050 (0.079)
Cutpoints ^a		
cut_2_1	−1.778 (0.145)***	
cut_2_2	−1.021 (0.139)***	
cut_2_3	0.286 (0.138)**	
cut_2_4	1.529 (0.139)***	
$\rho_{\varepsilon\theta}$		−0.396 (0.112)***
Observations		4,212

Table 2. Impact of happiness on maize productivity: CMP model estimates

Note(s): Maize productivity is measured at 1,000 kg/mu; ** < 0.05 and *** < 0.01. The reference region is East. Standard errors are presented in parentheses

^aCutpoints in ordered probit model estimation indicate where the latent variable is cut to create the five happiness groups in the data

3.1 Determinants of farmers’ happiness

First, let us consider how different factors affect farmers’ happiness – the results are presented in the second column of Table 2. They show that household heads’ happiness is affected by their age and education, farm size, machinery, subsidy, access to credit, car and location dummies. For example, we find that age is positively associated with farmers’ happiness. Using the World Values Survey data for 95 economies, Frijters and Beaton (2012) also found that older people are happier. The significant and positive coefficient of the machinery variable suggests that owning machinery increases farmers’ happiness. Machines help farmers save time and effort allocated to toilsome farm work, leaving them with more time for leisurely pursuits. Although the Pareto inefficiency of farm subsidies has received considerable attention (Gul and Pesendorfer, 2007), whether subsidies enhance farmers’ happiness remains largely unaddressed in the literature. We show that subsidies improve farmers’ happiness. Receiving subsidies alleviates farmers’ financial burden while helping them accumulate savings, invest in their futures and cope with crop failures and price volatility.

Interestingly, having access to credit appears to compromise happiness. Considering that credit access may allow households to increase consumption (Kollamparambil, 2020) and facilitate farm and business investments that generate cash flows (Devoto et al., 2012), this may seem counterintuitive. However, by enabling households to assume larger debt, credit may undermine happiness (Liu et al., 2020). Our results lend credence to this reasoning. Car ownership positively affects farmers’ happiness. Only a small proportion of Chinese rural households own a car. Car ownership is a symbol of affluence and may impart a perception of

relative prosperity, thereby contributing to an increase in farmers' happiness. The estimation of location-specific variables shows that farmers in China's central and western regions are less happier relative to those in the eastern region. We find that farmers who donate to others report greater happiness than farmers who do not, a finding that is largely supported by findings of previous studies (Aknin and Whillans, 2021).

3.2 Determinants of maize productivity

The last column of Table 2 reports the factors affecting maize productivity. Turning to the variable of our primary interest, we show that farmers' happiness significantly and positively affects maize productivity. We provide two explanations here. First, farmers with a positive mental condition would make on-farm decisions like "when and how to apply fertilizers and pesticides" with careful considerations. This helps tackle uncertainty and potential risk in maize production and enables farmers to achieve higher productivity. In contrast, depression and anxiety may result in irrational and unanalytical decisions and cause productivity losses. Second, farmers who experience positive emotions can work more effectively. Their communication and management also improve (Isham *et al.*, 2021). Effective farm management contributes to higher maize output. According to the World Happiness Survey in 2020, the happiness level among the Chinese ranked 52nd (Helliwell *et al.*, 2021). Our findings highlight the potential benefits of improving Chinese farmers' happiness to enhance maize productivity.

To check whether the positive relationship between farmers' happiness and productivity holds for other crops, we also estimate the impact of farmers' happiness on wheat and rice productivity. To save space, we only present the coefficients of the happiness variable. The results (see Table A2 in the Appendix) show that farmers' happiness is positively associated with wheat and rice productivity. Thus, these findings verify the robustness of the positive association between farmers' happiness and farm productivity – the results hold for each of the three crops.

The estimates in Table 2 show that maize productivity is also affected by other factors. For example, age negatively impacts productivity, suggesting that older farmers have a relatively low productivity level. Working capacity decreases with age, and thus, younger farmers are more productive than older ones. The household size has a significant and negative impact on maize productivity. A larger household size usually means more children and older members (i.e. a higher dependency ratio) to look after, which may reduce the time and money allocated to farm production. Unsurprisingly, machinery ownership significantly improves maize productivity. The productivity-increasing effect of farm machinery use has been reported in previous studies (Ma *et al.*, 2018; Zhou *et al.*, 2020). Access to credit has contrasting effects on happiness and productivity – it compromises happiness but improves productivity. A possible reason is that farmers who have access to credit may accumulate

Happiness level	Maize productivity (predicted)	Marginal return in maize productivity	95% confidence intervals	
Happiness = 1	0.036 (0.029)***		0.029	0.043
Happiness = 2	0.113 (0.027)***	0.077	0.110	0.117
Happiness = 3	0.197 (0.030)***	0.084	0.195	0.198
Happiness = 4	0.275 (0.029)***	0.078	0.274	0.277
Happiness = 5	0.358 (0.030)***	0.083	0.356	0.361

Note(s): Marginal return in maize productivity refers to the difference in maize productivity between a higher level and the lowest happiness level (the level of happiness = 1)

*** < 0.01

Table 3.
Distributions of
predicted maize
productivity by
happiness levels

debt and consequently experience psychological stress and unhappiness. However, credit allows farmers to invest in initiatives such as acquiring fertilizers and machinery that generate cash flows and enhance productivity (Devoto *et al.*, 2012; Narayanan, 2016). Finally, our estimates show that farmers in the central region are more productive than those in the eastern region.

Having ascertained a positive and significant effect of happiness on maize farmers' productivity, we predict productivity based on the coefficients estimated in Table 2 and compute the specific marginal increases in productivity associated with greater happiness. The results (see column 2 of Table 3) show that maize productivity is monotonically increasing with farmers' self-reported happiness levels. Nevertheless, the marginal return in maize productivity changes unevenly (see column 3 of Table 3). For example, when farmers' happiness levels increase from 3 to 4, the marginal return in maize productivity is 0.078. This corresponds to an increase in maize productivity by 40%. However, the marginal return in maize productivity is 0.083 when farmers' happiness levels increase from 4 to 5, corresponding to an increase in maize productivity by 30%.

4. Conclusion

The nexus between happiness and productivity is mainly examined for firms, but it has been neglected for farms. This paper provided the first analysis of the relationship between farmers' happiness and farm productivity, taking maize production in China as an example. The CMP model was employed to account for the endogeneity that inevitably arises in regression models studying happiness. The results showed that farmers' happiness has a significant and positive impact on maize productivity. This result also holds rice and wheat productivities. Although monotonic, the marginal return in maize productivity changes unevenly with increments in farmers' self-reported happiness on a five-point Likert scale. These findings confirm the positive relationship between happiness and productivity at the farm level. As such, our findings emphasize the importance of improving farmers' subjective well-being to achieve higher farm productivity.

The results have implications for China and other developing countries aiming for sustainable agricultural development. Considering that happiness and productivity are positively related, policy interventions that lead to greater happiness may engender higher productivity. For example, subsidies and happiness are positively related. Thus, governments can improve farmers' productivity through carefully designed subsidy programs. We also find that ownership of machinery and cars, which symbolize household wealth and help farmers save time and effort, is associated with greater happiness. Targeted loans and initiatives to help farmers diversify their incomes, allowing them to acquire such goods would help improve farmers' happiness and thus their productivity.

Furthermore, farm size increases happiness. This suggests that promoting rural land transfer policies that help farmers increase the size and scale of their farming operations could improve farmers' happiness. Nevertheless, albeit insignificantly, farm size decreases farm productivity. This underscores the importance of providing training and education to farmers – larger farms may not lead to higher productivity in the absence of requisite training.

This study focuses only on the happiness of household heads. We recognize that the happiness of other farmers, such as hired laborers, may also affect farm productivity. We suggest that future research be devoted to examining farm productivity more exhaustively, accounting for the happiness of hired laborers and the household as a whole. Besides, we have focussed on maize production in this study. Future studies should consider investigating how the happiness of farmers affects the productivity of other crops such as soybean.

Note

1. The latest 2017 CHFS is available; however, it does not contain information on crop yields.

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Appendix

Table A1. Validity test of the employed IV	Instrumental variable	Dependent variables	Statistics	p-value
	<i>Pearson correlation analysis</i>			
	Donation (IV)	Happiness	Correlation = 0.055***	0.000
	Donation (IV)	Maize productivity	Correlation = 0.018	0.234
	<i>Falsification test</i>			
	Donation (IV)	Happiness	Chi2(1) = 7.86***	0.005
	Donation (IV)	Maize productivity	F(1,4196) = 2.45	0.1175
	Note(s): *** < 0.01. p-values are presented in parentheses			

Table A2. Impacts of happiness on rice and wheat productivities: second- stage estimation of the CMP model	Variables	Rice productivity	Wheat productivity
	Happiness	0.052 (0.026)**	0.047 (0.021)**
	Control variables	Yes	Yes
	Observations	2,254	1,976
	Note(s): ** < 0.05. Standard errors are presented in parentheses		

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